CTSCleaner (CleanConnect)

A marketplace platform connecting customers with individual cleaners and cleaning companies. AI-powered room analysis determines fair pricing, with built-in negotiation, escrow payments, KYC verification, and email OTP validation.

Live: https://ctscleaner.vercel.app

Tech Stack

Layer	Technology
Framework	Next.js 16 (App Router)
Language	TypeScript 5
Styling	Tailwind CSS 4
Database	Supabase (PostgreSQL)
Storage	Supabase Storage (room photos)
AI Vision	OpenAI gpt-4o-mini / gpt-4o
Email	Resend
Deployment	Vercel (project: ctscleaner)

User Journeys

There are two top-level paths on the platform:

1. Post a Job (Customer Side)

Anyone needing cleaning services — individual or business.

- Individual — posts for home/personal property. No VAT applied.
- Business — posts for commercial property. UK VAT (20%) auto-calculated. Business domain email required (no Gmail/Hotmail/Yahoo etc.).

2. Service Provider (Cleaner Side)

Anyone offering cleaning services — individual or company.

- Individual Cleaner — any email. One-off onboarding fee: £99 (non-refundable).
- Cleaning Company — business domain email required. One-off onboarding fee: £179 (non-refundable).

Features

Post a Job

- Step 1: Choose poster type — Individual or Business
- Step 2: Select number of rooms (Kitchen, Living Room, Bedroom, Play Room, Toilet/Bathroom)
- Step 3: Upload room photos — AI analyses each one (dirtiness, clutter, hours, price, risk)
- Step 4: Price summary with platform fee, optional content insurance, and VAT (business only)
- Step 5: Contact details — business email enforced for business accounts
- Funds held in escrow, released 24 hours after job completion — no automatic release

AI Room Analysis

- Images compressed to 768px max width, JPEG 0.65 quality before upload
- Uploaded to Supabase Storage, then sent to OpenAI Vision
- Returns: room type, cleanliness score, clutter score, estimated hours, risk level, suggested price, recommended cleaners
- Model: gpt-4o-mini by default (cheap, fast)
- Auto-switches to gpt-4o when declared content value exceeds £10,000
- Model badge shown on each analysis result

Service Provider Registration

- Email OTP verification via Resend (6-digit code, 10-minute expiry)
- Business companies must use a business domain email (validated on both card selection and email field)
- Pre-agree to auto-accept 90% price tier at registration
- KYC required every 6 months — business email NOT required for KYC

Negotiation System (4-Price Tiers)

- When a customer selects a cleaner, they enter rooms and upload photos
- AI determines the base price
- Customer offered 4 tiers: 100%, 95%, 90% (default), 80%
- Email sent to cleaner to accept
- If cleaner pre-agreed to 90% → auto-accepted instantly, customer proceeds to payment

UK VAT (Business Accounts)

- Standard rate: 20%
- Applied to: service subtotal + platform fee
- VAT receipt issued on payment
- VAT number field (optional) collected at job posting

Content Insurance

- Optional add-on when posting a job
- 8% flat rate of declared content value
- Maximum cover: £4,000
- Premium capped at £320

Pricing Model

Component	Rate
Base hourly rate	£20/hr
Clean room (score 1–3)	Minimum 2 hours
Moderate dirt (score 4–6)	+50% hours
Heavy/Extreme dirt (score 7–10)	Minimum 4 hours
Platform fee (customer)	5%
Matching fee (cleaner payout)	7.5%
UK VAT (business accounts only)	20%

Registration Fees (one-off, non-refundable)

Type	Fee
Individual cleaner	£99
Cleaning company	£179

Navigation Structure

Nav Item	Destination	Who uses it
Home	/	Everyone
Dashboard	/dashboard	Admin
Find Cleaners	/cleaners	Customers
Post a Job	/post-job	Individual & Business customers
Service Provider	/register-cleaner	Individual cleaners & companies

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Project Structure

```
src/
├── app/
│   ├── api/
│   │   ├── analyze-room/     # OpenAI Vision room analysis
│   │   ├── send-otp/         # Resend email OTP (6-digit, 10min expiry)
│   │   └── verify-otp/       # OTP verification
│   ├── cleaners/
│   │   ├── page.tsx          # Browse all cleaners
│   │   └── [id]/page.tsx     # Cleaner profile + 4-price booking flow
│   ├── dashboard/page.tsx    # Admin overview
│   ├── jobs/[id]/page.tsx    # Job detail page
│   ├── post-job/page.tsx     # Post a cleaning job (individual or business)
│   ├── register-cleaner/     # Service Provider registration + OTP flow
│   ├── register-individual/  # Landing: post job or register as cleaner
│   ├── register-business/    # Landing: post job or register as company
│   ├── privacy/              # Privacy policy
│   ├── terms/                # Terms of service
│   └── support/              # Support page
├── components/
│   ├── PostJobSection.tsx    # Full job posting form with AI pricing + VAT
│   ├── RegisterBusinessSection.tsx
│   ├── FeeStructure.tsx
│   ├── HeroSection.tsx
│   ├── HowItWorks.tsx
│   ├── JobListings.tsx
│   ├── Navbar.tsx            # Post a Job + Service Provider
│   └── Footer.tsx
├── context/
│   └── CleanConnectContext.tsx  # Supabase CRUD (jobs, cleaners, businesses)
└── lib/
    ├── supabase.ts           # Supabase client
    ├── analysisLimit.ts      # Daily free analysis limit (5/day)
    ├── sampleCleaners.ts     # 7 placeholder cleaners (2 individual, 2 company, 3 service provi
    └── sampleJobs.ts         # Sample jobs for demo
```

Placeholder Data

Individual Cleaners

Name	City	Rate	Auto-Accept 90%
Sarah Mitchell	London (SE1)	£20/hr	Yes
James Okafor	London (E1)	£25/hr	No

Cleaning Companies

Name	City	Rate	Auto-Accept 90%
Sparkle Pro Clean Ltd	Manchester	£30/hr	Yes
Clean & Shine Services Ltd	London	£28/hr	No

Service Providers

Name	Type	City	Rate
PrimeServ Facilities Ltd	Business	Birmingham	£32/hr
Priya Sharma	Individual	Birmingham	£20/hr
Tom Bradley	Individual	Leeds	£20/hr

Database Schema (Supabase)

`jobs`

Column	Type	Description
id	uuid	Primary key
rooms	jsonb	Room types and counts
total_rooms	int	Total rooms
subtotal	numeric	AI-calculated subtotal
total	numeric	Total inc. platform fee, VAT, insurance
email	text	Customer email
phone	text	Customer phone
postcode	text	Job postcode
city	text	Job city
status	text	awaiting / accepted / completed
conditions	jsonb	Per-room condition map
created_at	timestamptz	Timestamp

`cleaners`

Column	Type	Description
id	uuid	Primary key

name	text	Full name or company name
email	text	Contact email (business domain for companies)
phone	text	Contact phone
postcode / city	text	Location
bio	text	Description
type	text	individual / company
rate	numeric	Hourly rate (£)
radius	int	Coverage radius (miles)
company_number	text	Companies House number
auto_accept_90	boolean	Pre-agreed 90% auto-accept
created_at	timestamptz	Timestamp

`businesses`

Column	Type	Description
id	uuid	Primary key
company_name	text	Registered company name
company_number	text	Companies House number
vat_number	text	VAT number (optional)
contact_name	text	Primary contact
email / phone	text	Business contact details
postcode / city	text	Location
service_type	text	Type of cleaning needed
created_at	timestamptz	Timestamp

`analyses`

Column	Type	Description
id	uuid	Primary key
job_id	uuid	FK → jobs
room_key / room_index	text / int	Which room
image_url	text	Supabase Storage public URL

room_type	text	Detected room type
cleanliness_score	int	1–10
clutter_score	int	1–10
estimated_cleaning_hours	numeric	AI estimate
risk_level	text	Low / Moderate / High / Extreme
suggested_price_gbp	numeric	AI price (£)
recommended_cleaners	int	Suggested cleaner count
reasoning	text	One-sentence AI summary
created_at	timestamptz	Timestamp

`otp_codes`

Column	Type	Description
id	uuid	Primary key
email	text	Target email address
code	text	6-digit code
purpose	text	cleaner / customer
verified	boolean	Whether code was used
expires_at	timestamptz	10-minute expiry
created_at	timestamptz	Timestamp

API Routes

`POST /api/analyze-room`

Analyses a room image using OpenAI Vision and stores the result in Supabase.

Body:

```
{
  "imageBase64": "...",
  "mediaType": "image/jpeg",
  "roomKey": "kitchen",
  "roomIndex": 0,
  "jobId": "uuid (optional)",
  "useHighAccuracy": false
}
```

Response:

```
{
  "success": true,
  "analysis": {
    "room_type": "Kitchen",
    "cleanliness_score": 7,
    "clutter_score": 5,
    "estimated_cleaning_hours": 4,
    "risk_level": "High",
    "suggested_price_gbp": 80,
    "recommended_cleaners": 1,
    "reasoning": "Kitchen has grease buildup and dirty dishes."
  },
  "imageUrl": "https://..."
}
```

Model selection:

- useHighAccuracy: false → gpt-4o-mini (default)
- useHighAccuracy: true → gpt-4o (auto when content value > £10,000)

`POST /api/send-otp`
Sends a 6-digit verification code via Resend.

Body: `{ "email": "user@example.com", "purpose": "cleaner" }`

`POST /api/verify-otp`
Verifies a submitted OTP code.

Body: `{ "email": "user@example.com", "code": "123456" }`

Environment Variables

Variable	Used in	Description
`OPENAI_API_KEY`	Server	OpenAI Vision API
`NEXT_PUBLIC_SUPABASE_URL`	Client + Server	Supabase project URL
`NEXT_PUBLIC_SUPABASE_ANON_KEY`	Client + Server	Supabase anon key
`RESEND_API_KEY`	Server	Resend transactional email

All set in `.env.local` (local dev) and Vercel project `ctscleaner` (production).

Local Setup

```
git clone <repo>
cd CleanConnect
npm install

# Create .env.local
OPENAI_API_KEY=sk-proj-...
NEXT_PUBLIC_SUPABASE_URL=https://xshemmkygkuaynyjbqrx.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=eyJ...
RESEND_API_KEY=re_...

# Run Supabase schema (SQL Editor)
# paste supabase-schema.sql → run
# paste supabase-add-otp.sql → run (if otp_codes table missing)

npm run dev
```

Deployment

```
vercel --prod --scope olumide-s-projects1 --yes
```

Live at: **https://ctscleaner.vercel.app**

Business Rules Summary

Rule	Detail
Individual cleaner fee	£99 one-off, non-refundable
Company cleaner fee	£179 one-off, non-refundable
KYC renewal	Every 6 months
Business email for KYC	NOT required
Business email to register as company	Required
Business email to post as business	Required
Escrow release	24 hours after job completion
Auto-release	Never
Negotiation tiers	100%, 95%, 90% (default), 80%
Auto-accept tier	90% (if pre-agreed at registration)

Content insurance	8% of declared value, max £4,000
AI model (standard)	gpt-4o-mini
AI model (content > £10k)	gpt-4o
Image compression	768px max, JPEG 0.65 quality
UK VAT (business jobs)	20% on service + platform fee
Platform fee (customer)	5%
Matching fee (cleaner)	7.5% of payout